

# Motor Modeling

디지털제어 및 신호처리 - 2조

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유민결 유채은 김솔

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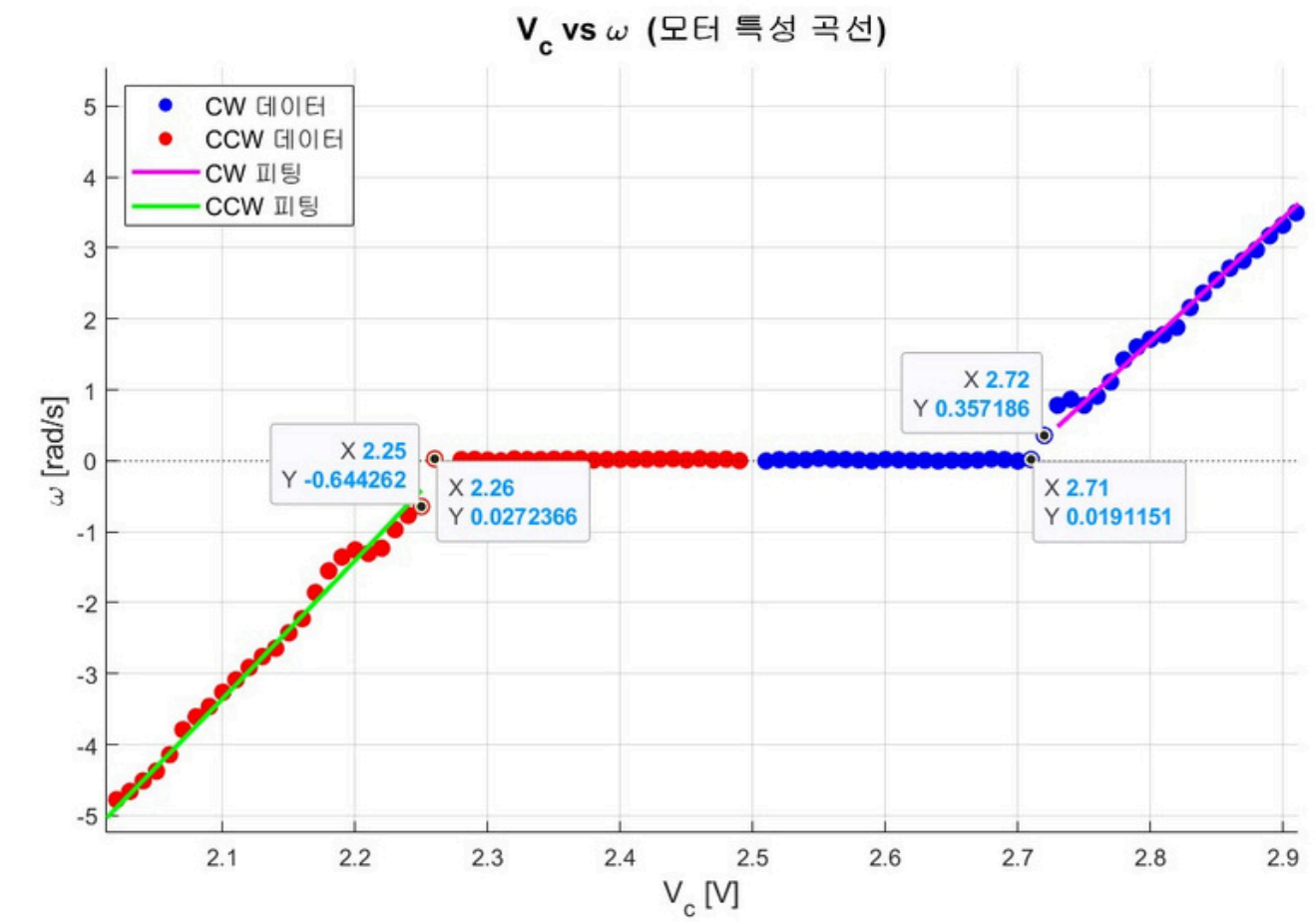
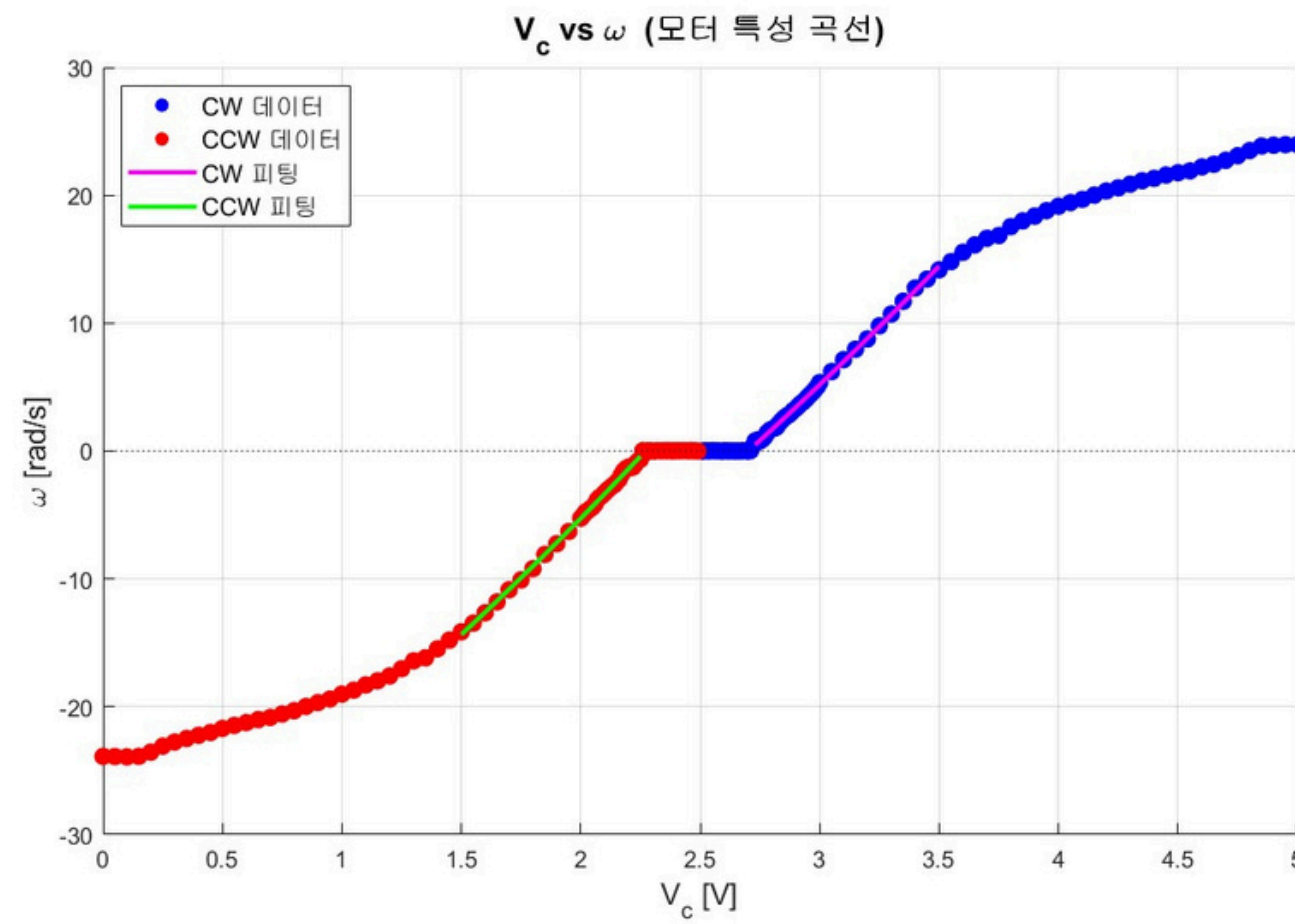
03. Validation

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# 1. Static Motor Modeling

## Deadzone 및 Saturation 확인

- $2 \leq V_c \leq 3$  : 0.01V 간격
- $V_c < 2$  or  $V_c > 3$  : 0.05V 간격으로 data 수집



2차로 fitting

# 1. Static Motor Modeling

## Linearization

- 선형화 범위 : 1.5 [V] (CCW)  $\leq V_{c\_sat} \leq$  3.5 [V] (CW)로 설정
- 입력전압 범위 : 전체 ( $-2.5 [V] \leq V_{cmd} \leq 2.5 [V]$ ) 사용한다고 설정

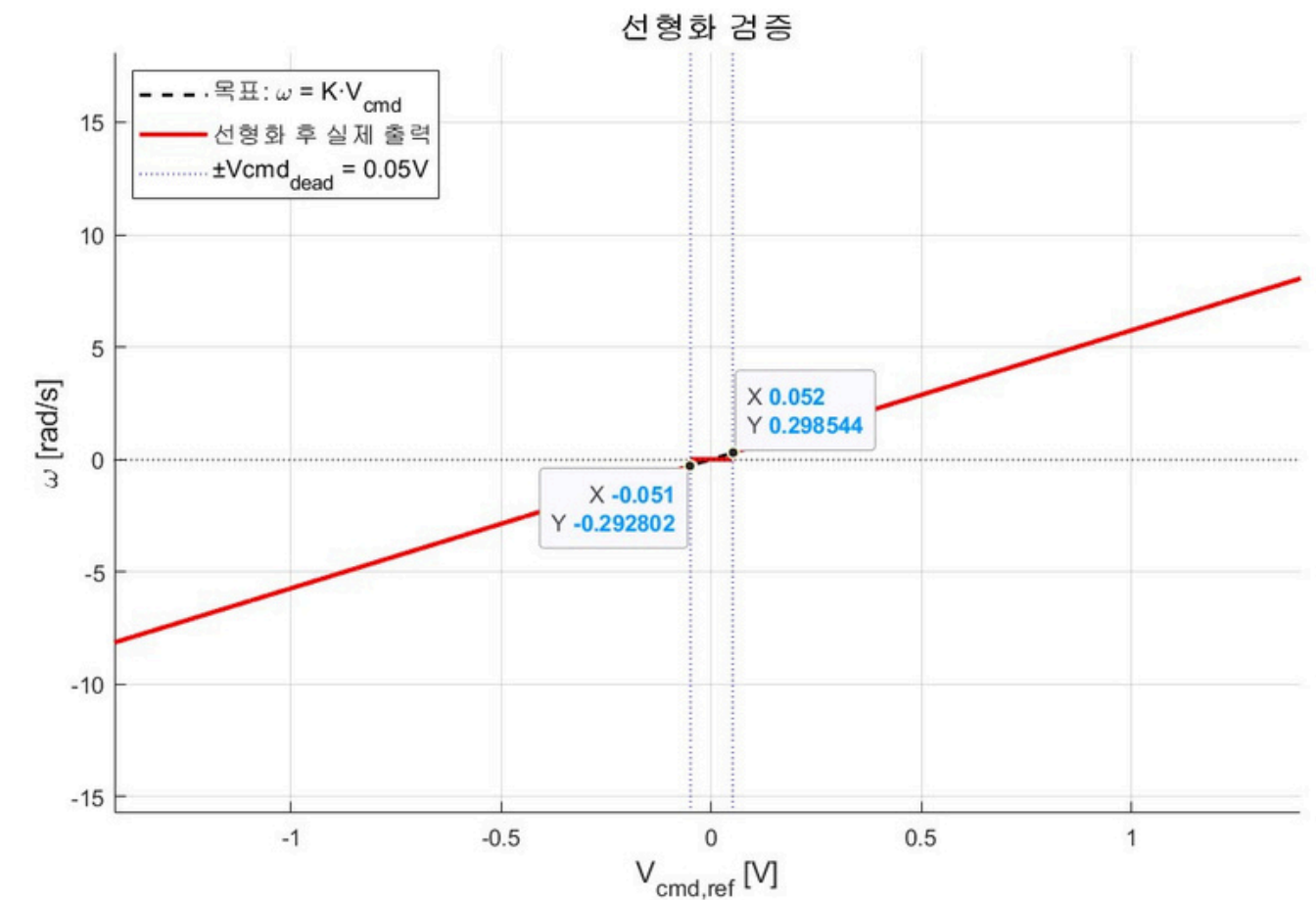
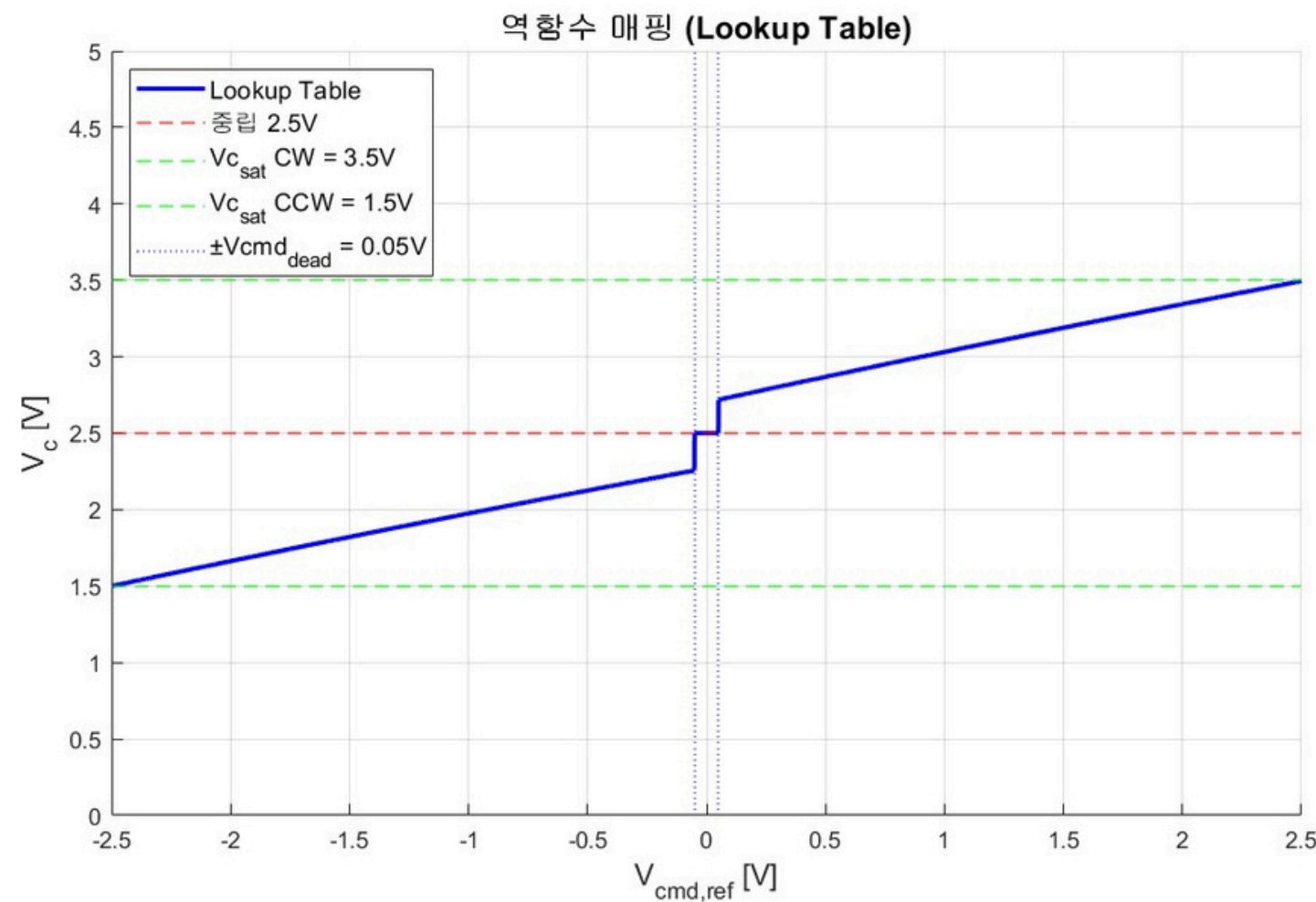
| Vcmd [V] | -2.5 | -2.0 | -1.5 | -1.0 | -0.5 | 0.0 | 0.5 | 1.0 | 1.5 | 2.0 | 2.5 |
|----------|------|------|------|------|------|-----|-----|-----|-----|-----|-----|
| Vc [V]   | 1.5  | 1.7  | 1.9  | 2.1  | 2.3  | 2.5 | 2.7 | 2.9 | 3.1 | 3.3 | 3.5 |

Mapping 관계

# 1. Static Motor Modeling

## Linearization

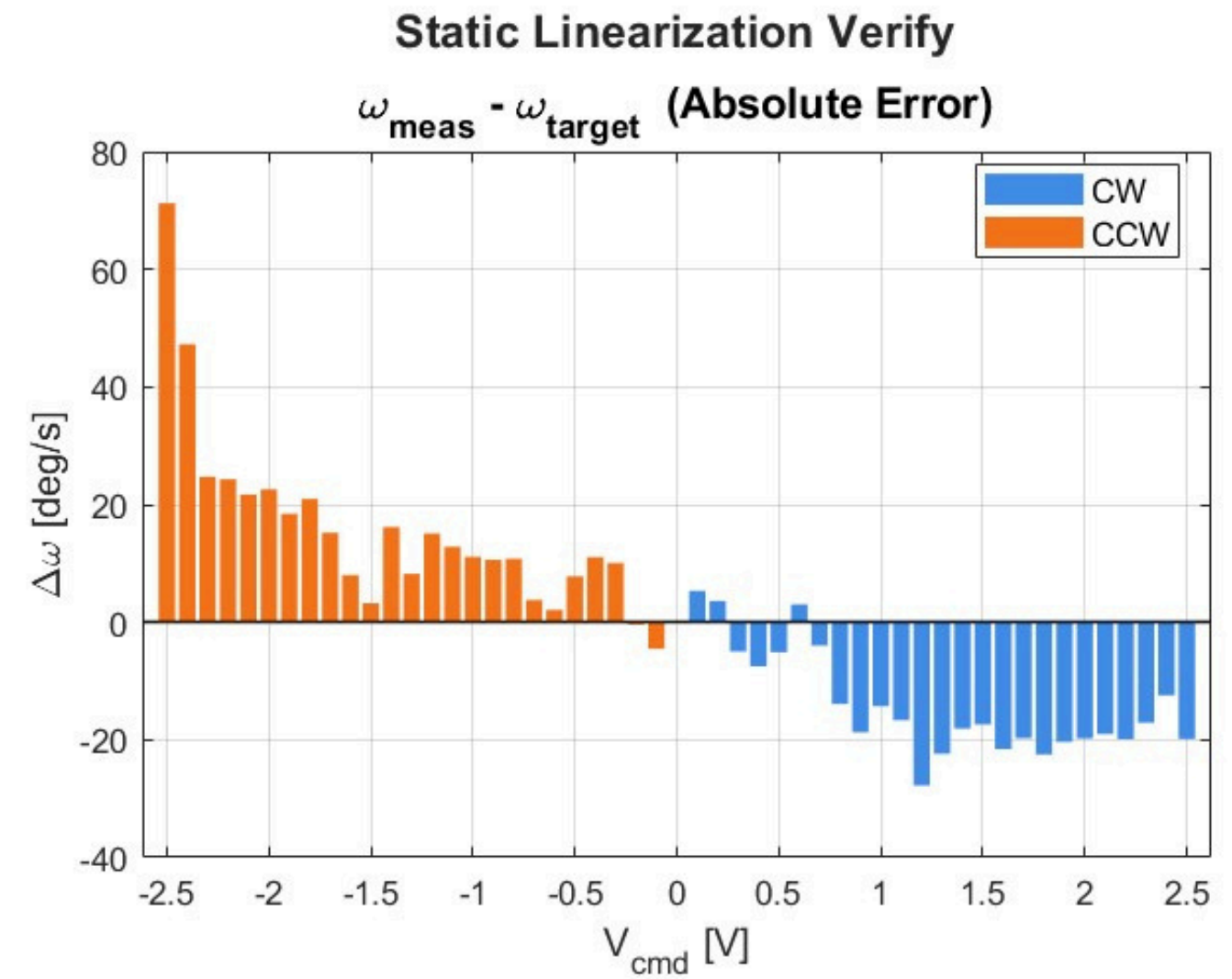
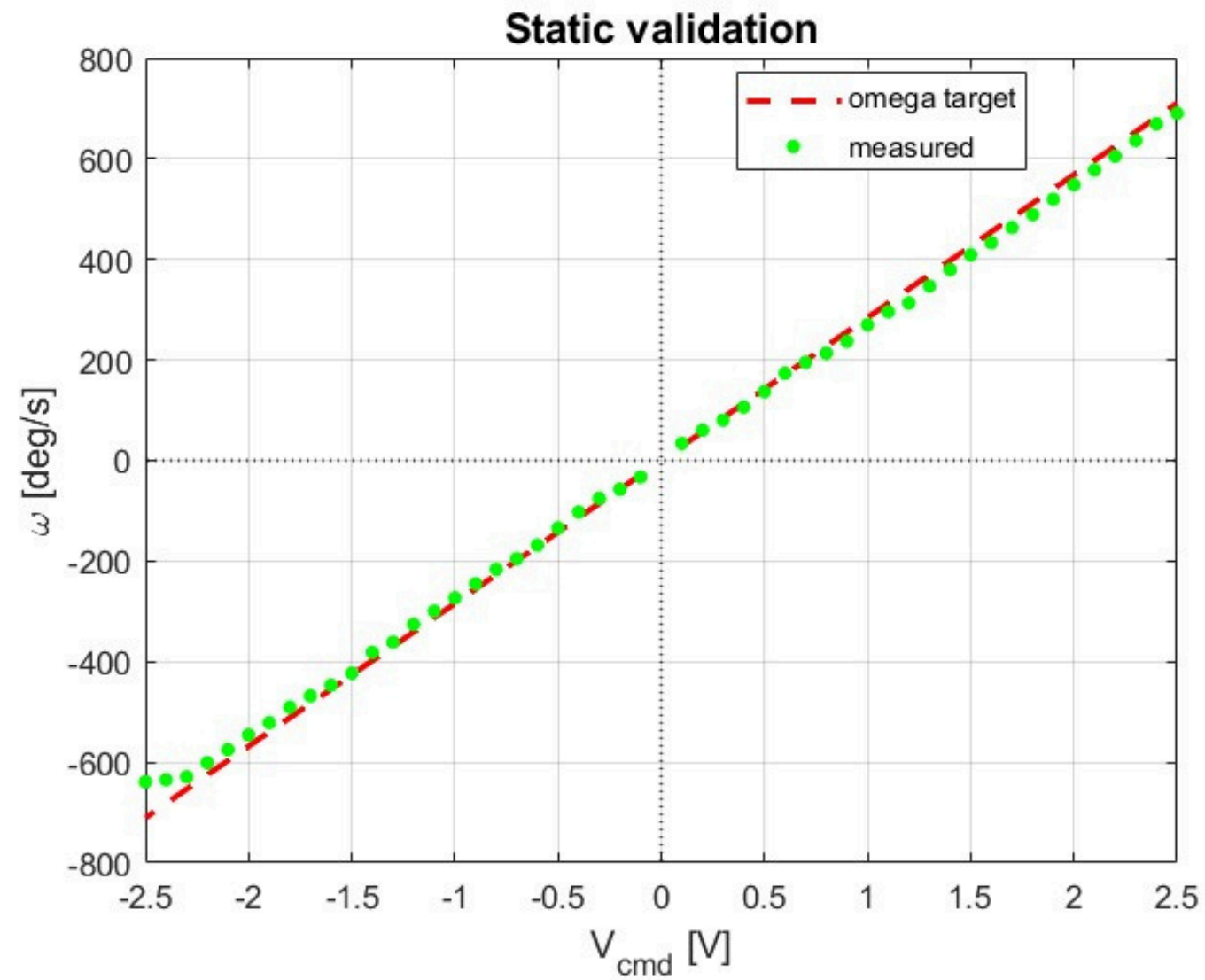
- 역함수 매핑 그래프



- $k = 5.7412 \text{ [(rad/s)/V]}$
- $V_{cmd}$ , Deadzone  
(CW) :  $0.052[V]$ , (CWW) :  $-0.051[V]$

# 1. Static Motor Modeling

## Static Validation



# 1. Static Motor Modeling

Slewrate Limit 확인 및 amplitude A의 구간 추정

$$5 \cdot V_1 \leq A \leq \frac{\text{slewrate limit}}{k \cdot \omega_{BW}}$$

$$5 \cdot 0.05 \leq A \leq \frac{\text{slewrate limit}}{4.96 \cdot \omega_{BW}}$$

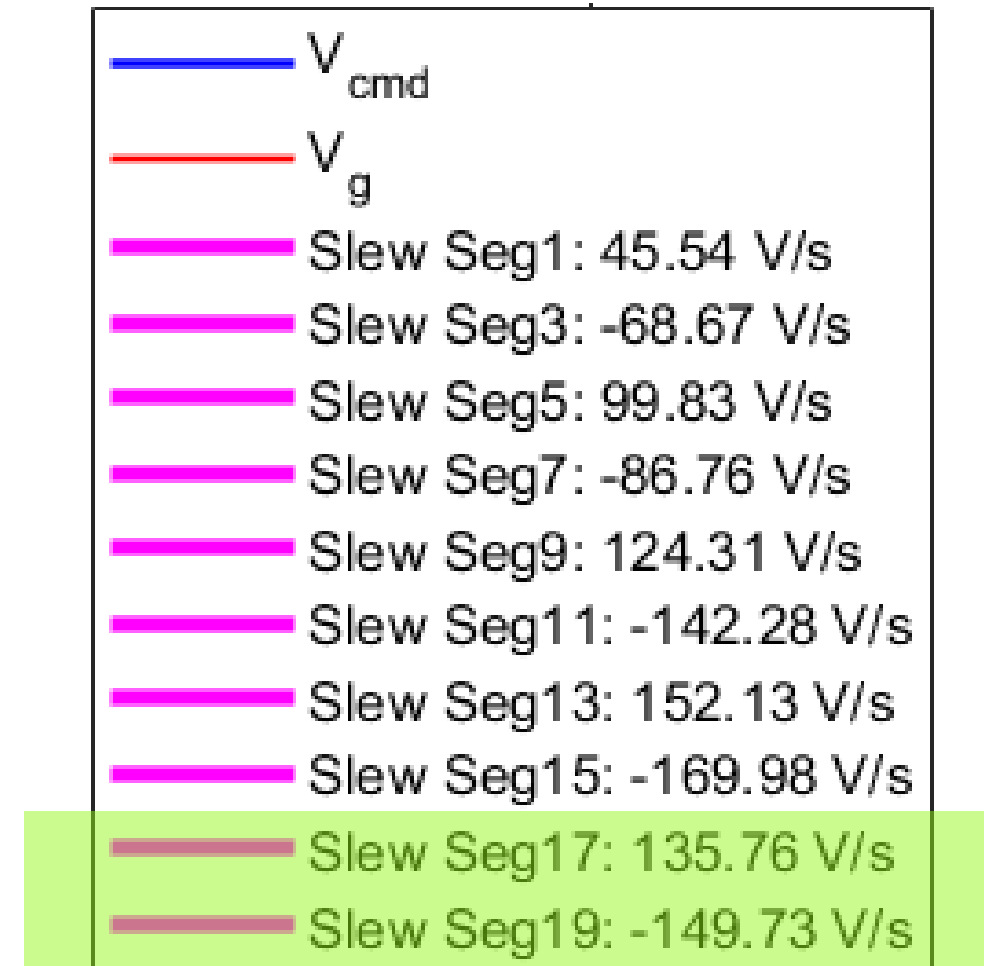
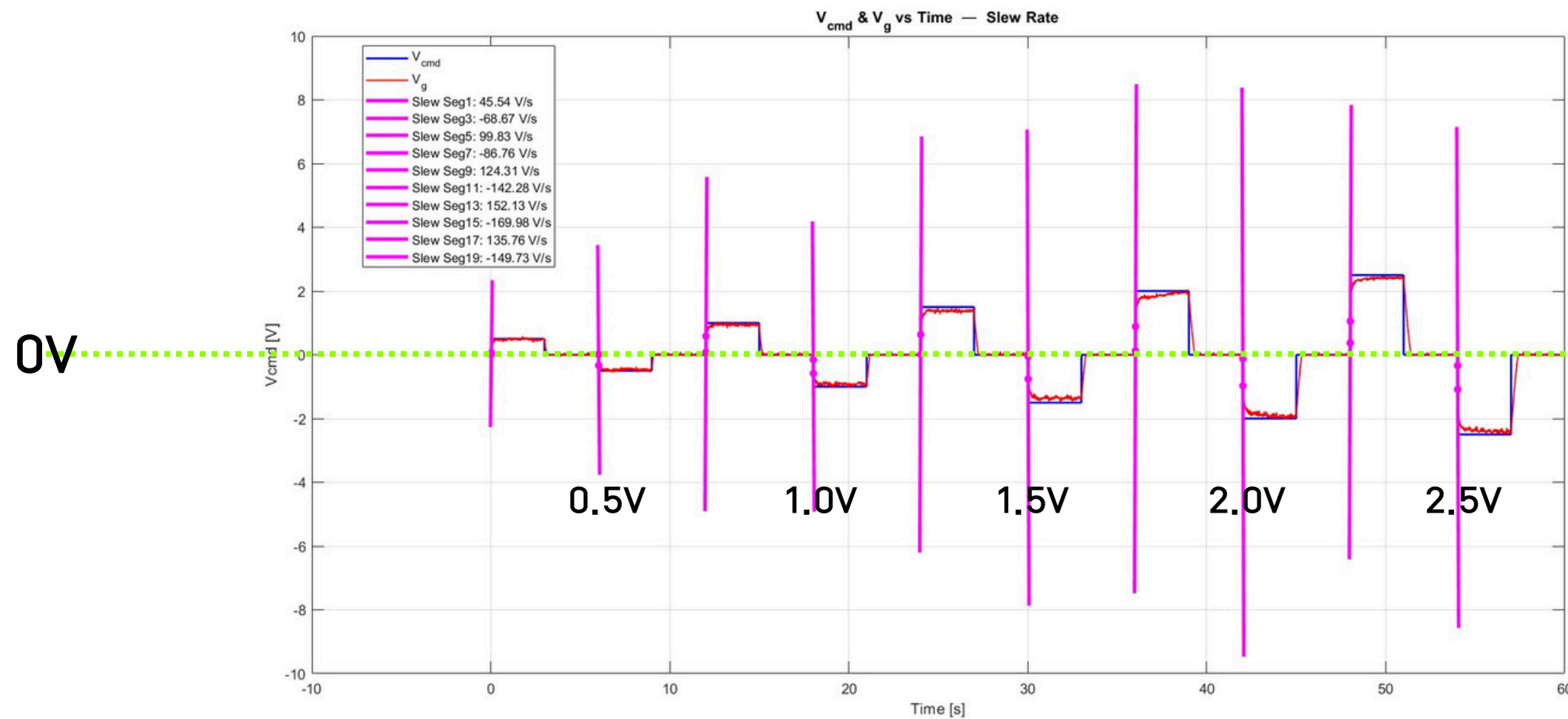
$$0.25 \leq A \leq \frac{\text{slewrate limit}}{4.96 \cdot \omega_{BW}}$$

A의 하한선 0.25 [V]

# 1. Static Motor Modeling

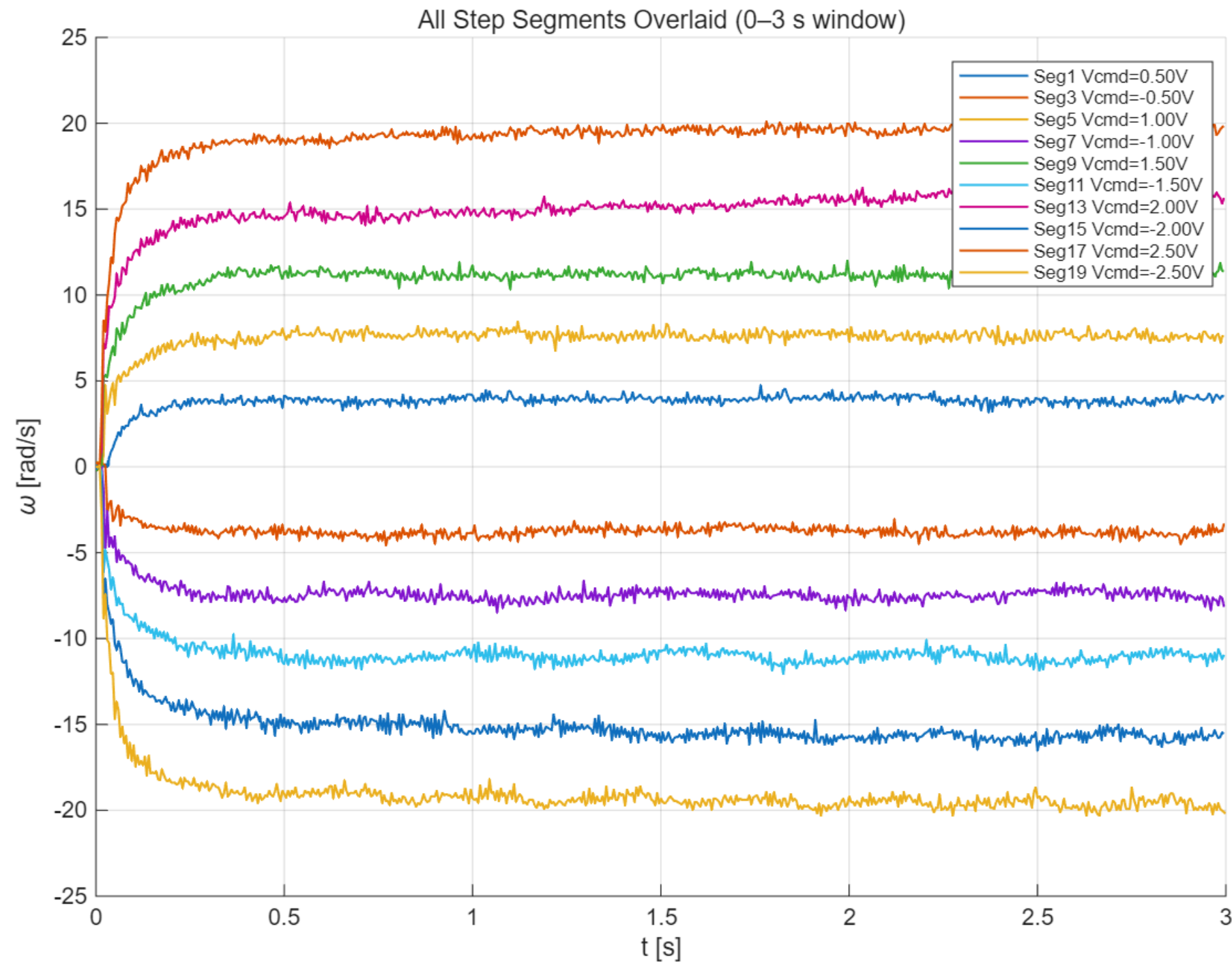
## Slewrate Limit 확인

- 최대 입력전압 2.5V까지도 큰 왜곡 없이 추종
- 2.5V로 넘어가는 구간에서 기울기값이 약간 감소



# 1. Static Motor Modeling

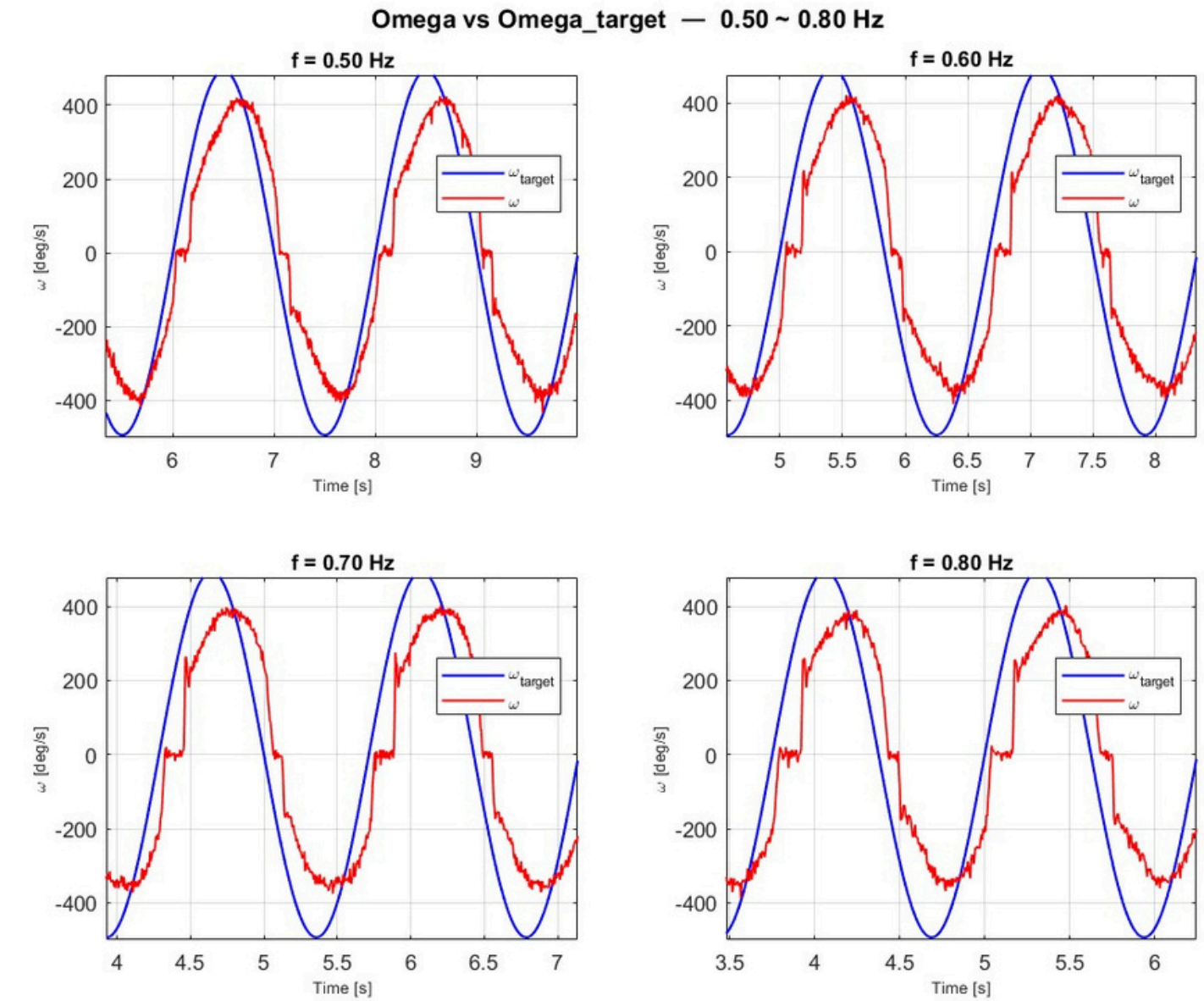
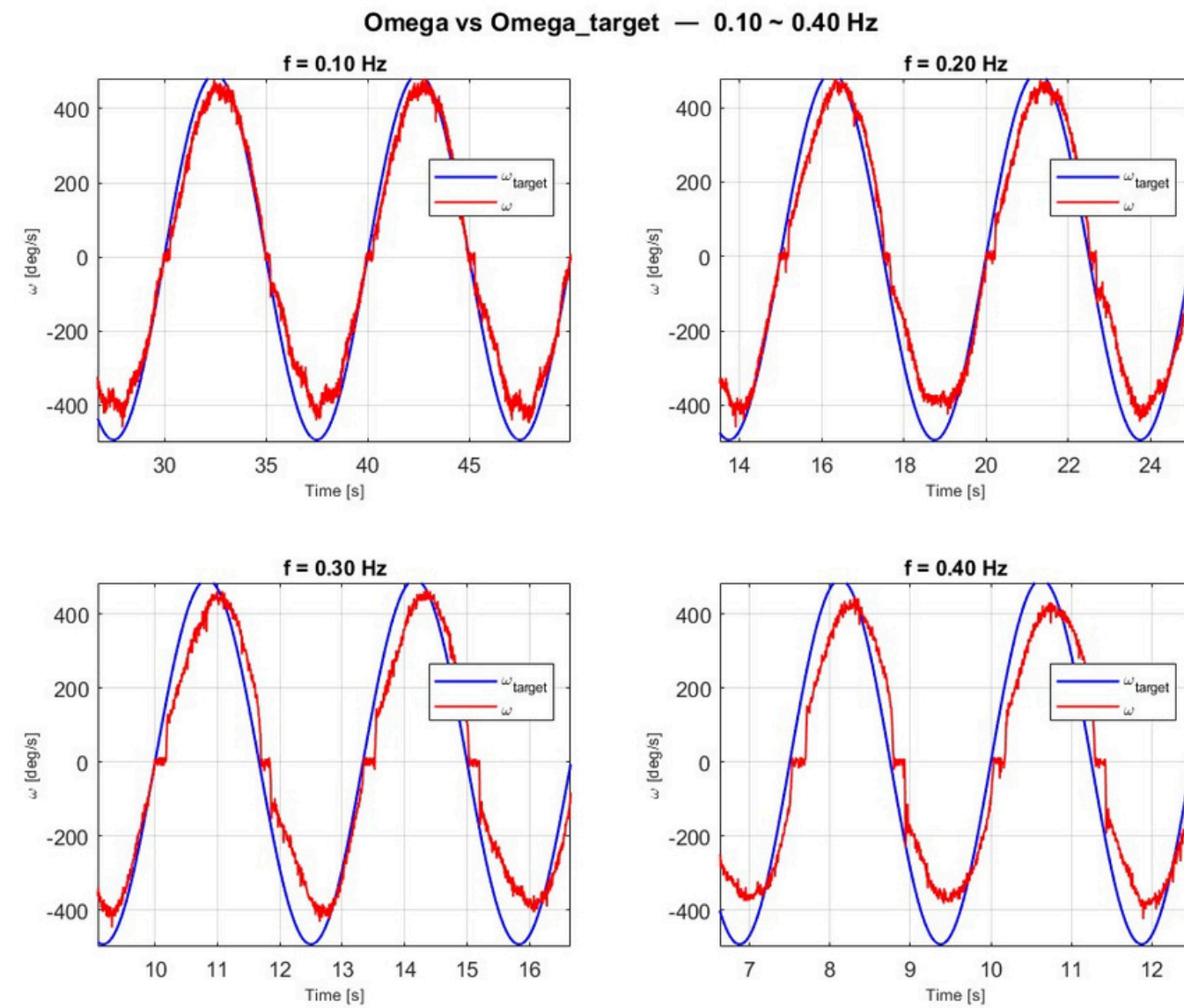
## Slewrate Limit 확인



모든 구간에서의 그래프를 한번에 plot  
기울기 간 명확한 차이 보기 어려움  
→ slewrate limit이 있다고 보기 어려움

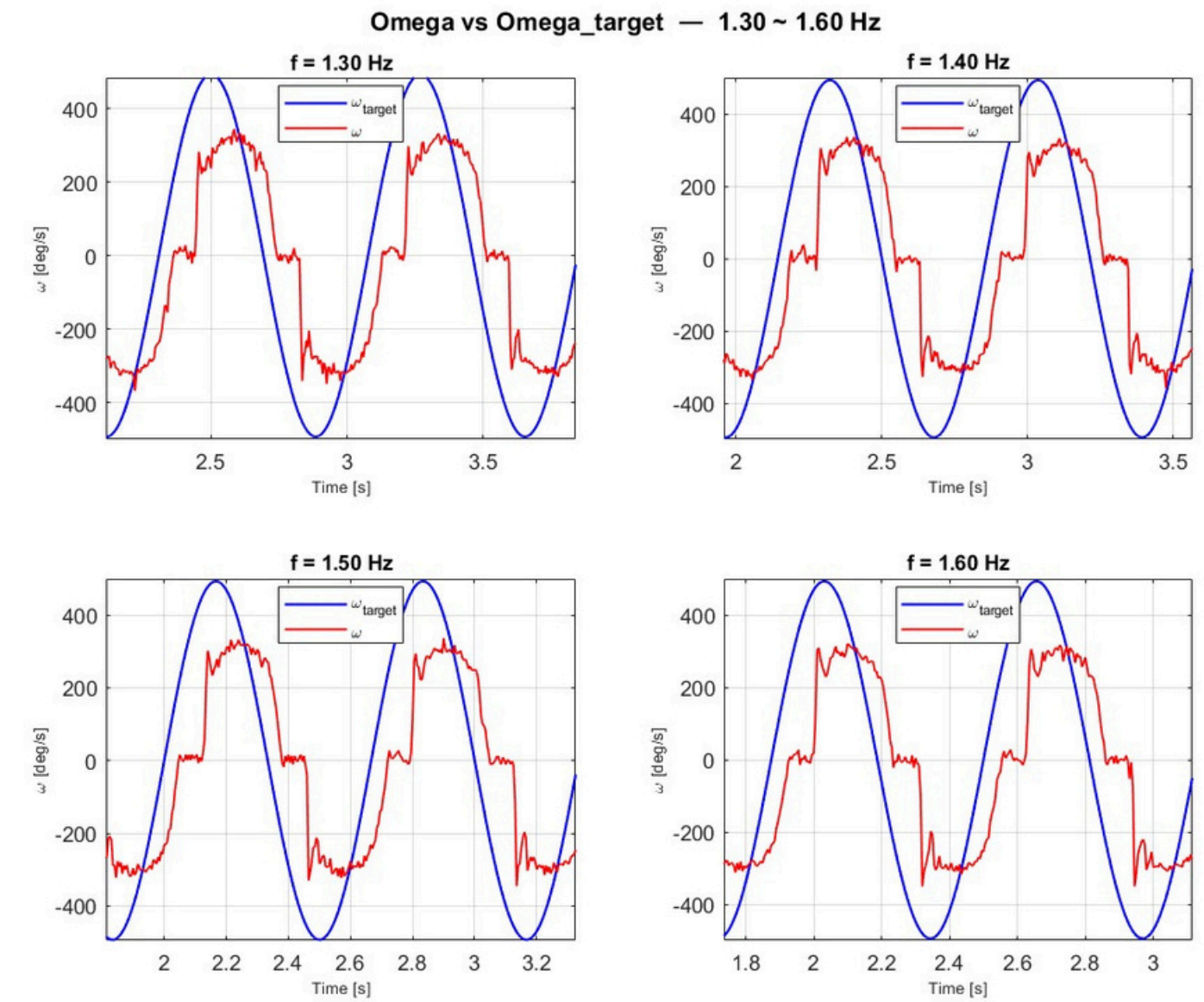
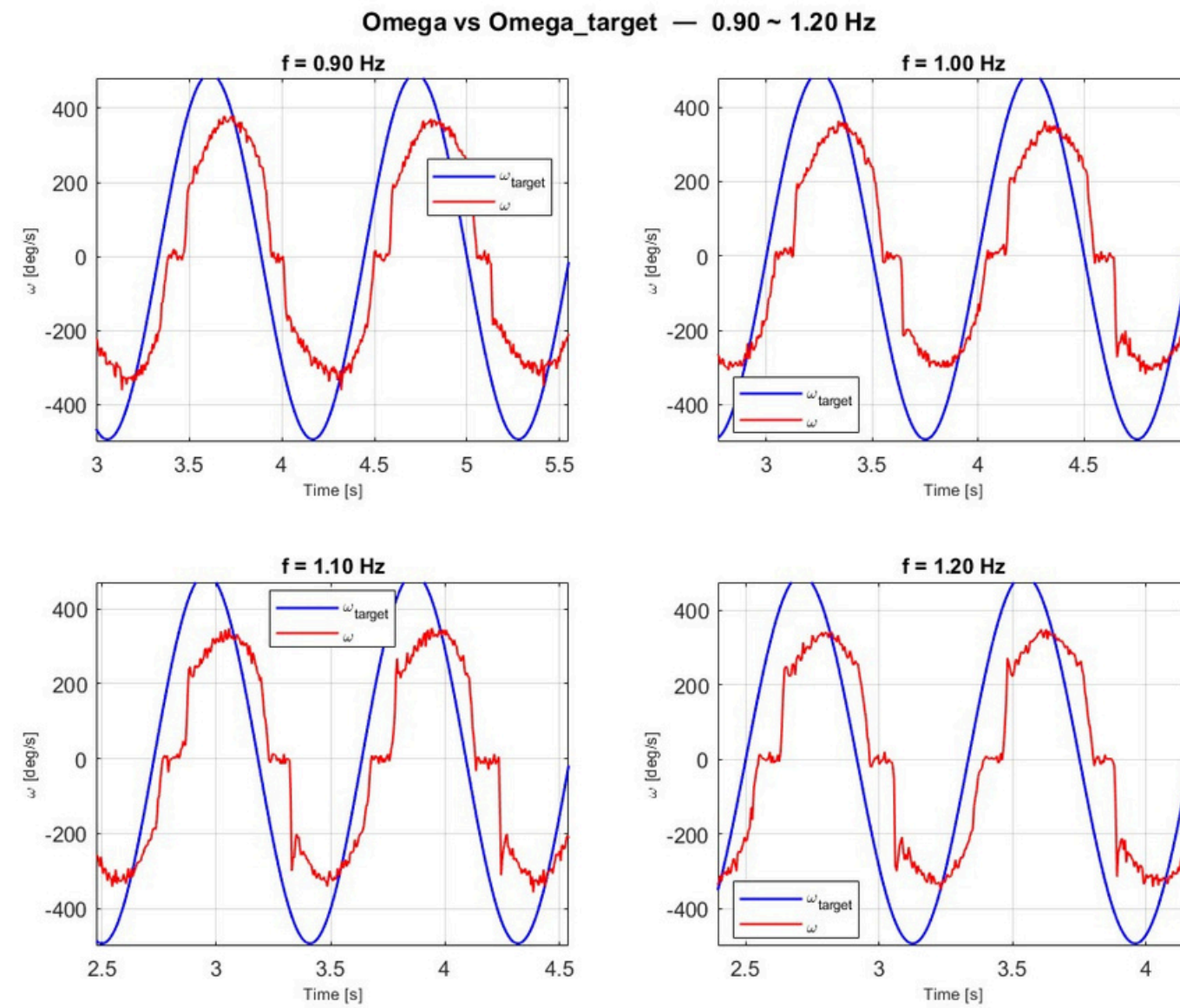
## 2. Dynamic Motor Modeling

### Frequency sweep



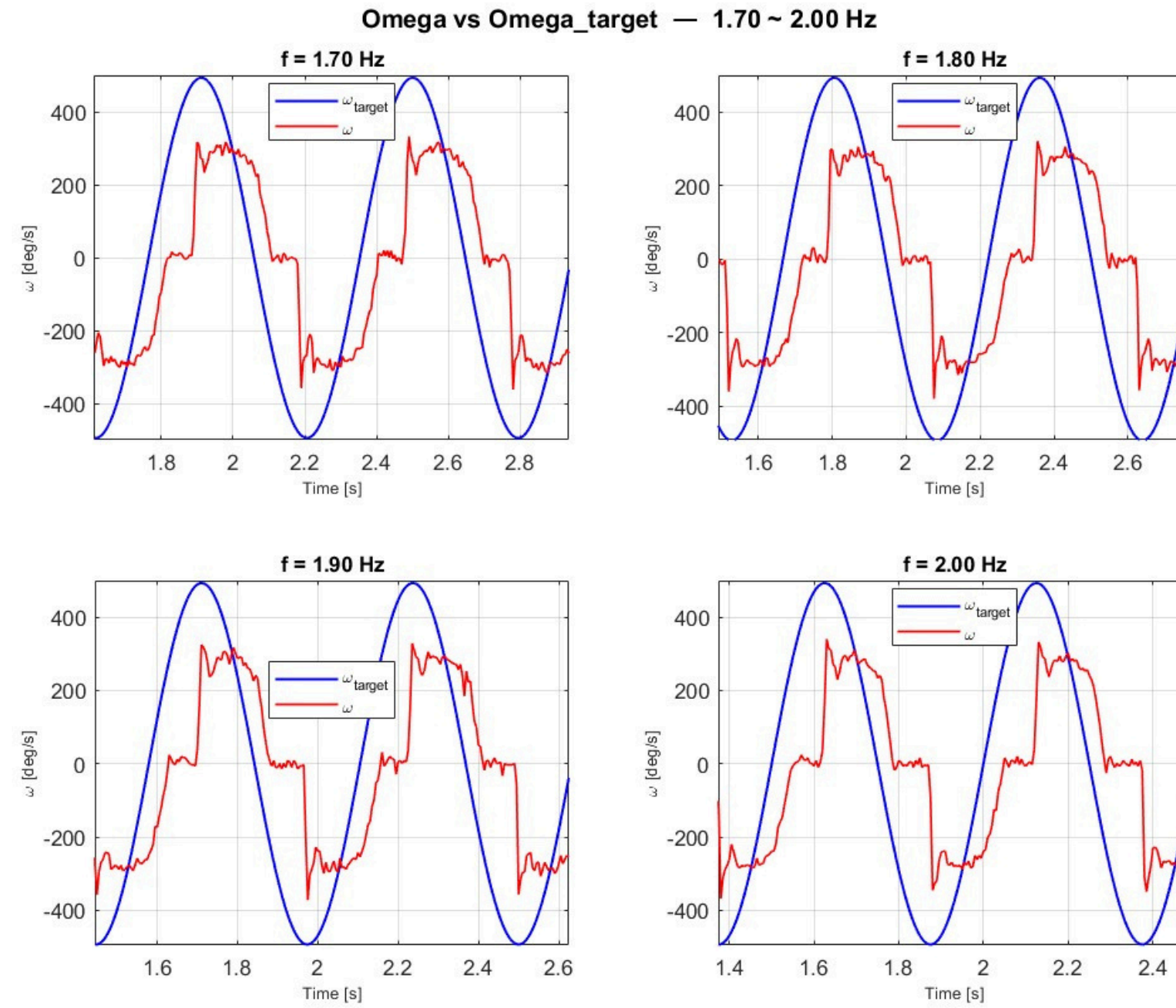
## 2. Dynamic Motor Modeling

### Frequency sweep



## 2. Dynamic Motor Modeling

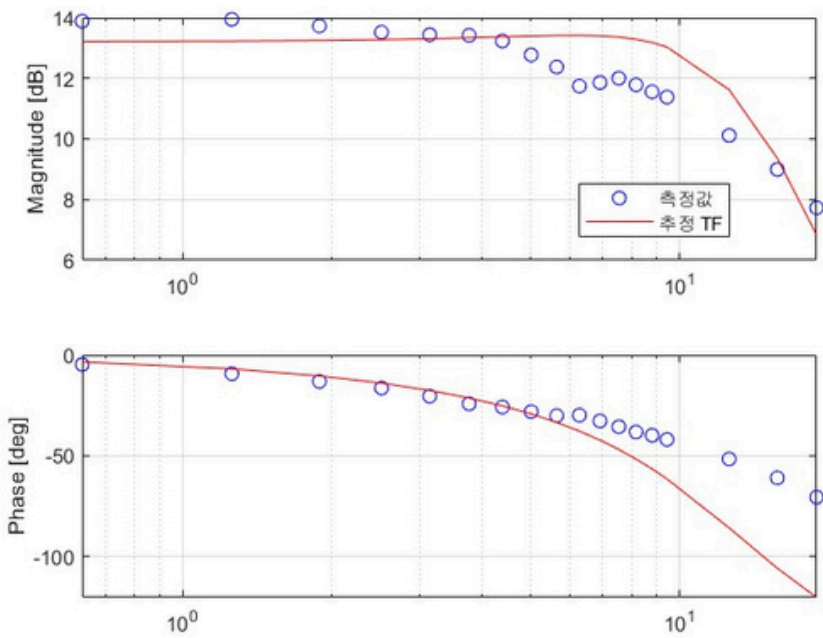
### Frequency sweep



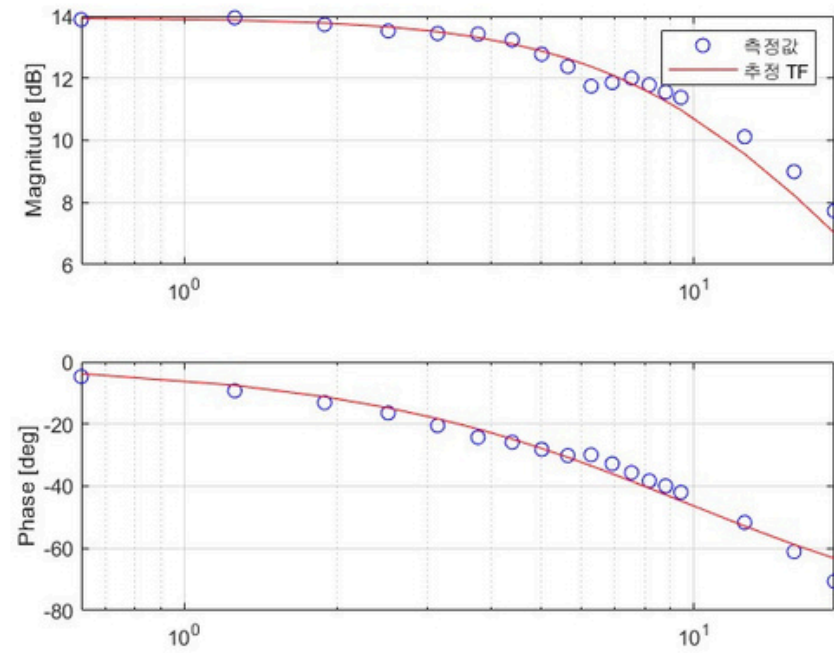
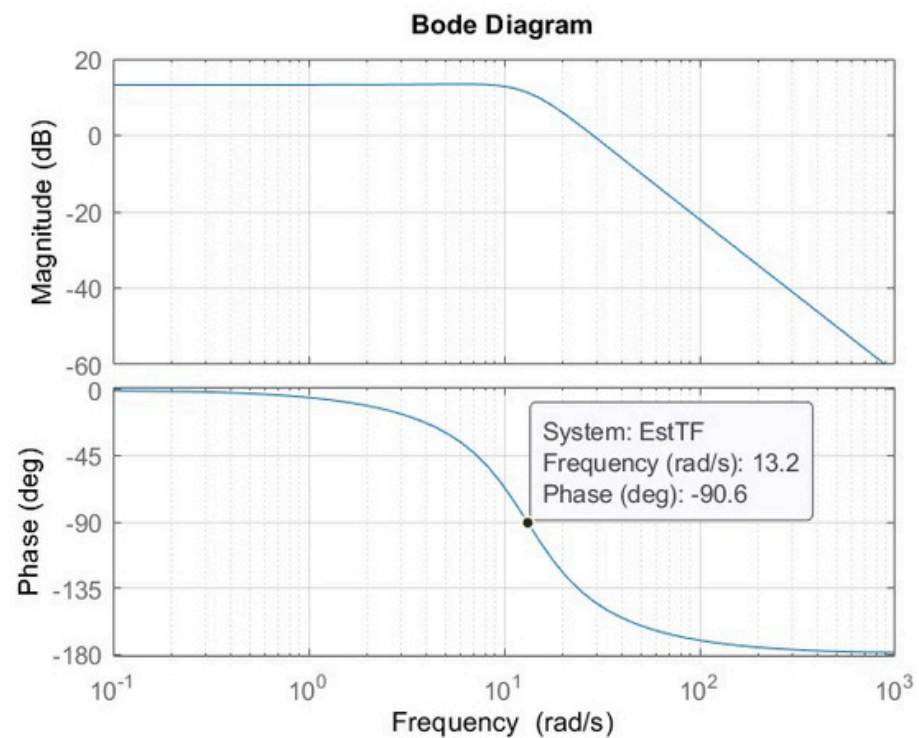
## 2. Dynamic Motor Modeling

Frequency sweep 및 모터 전달함수 추정

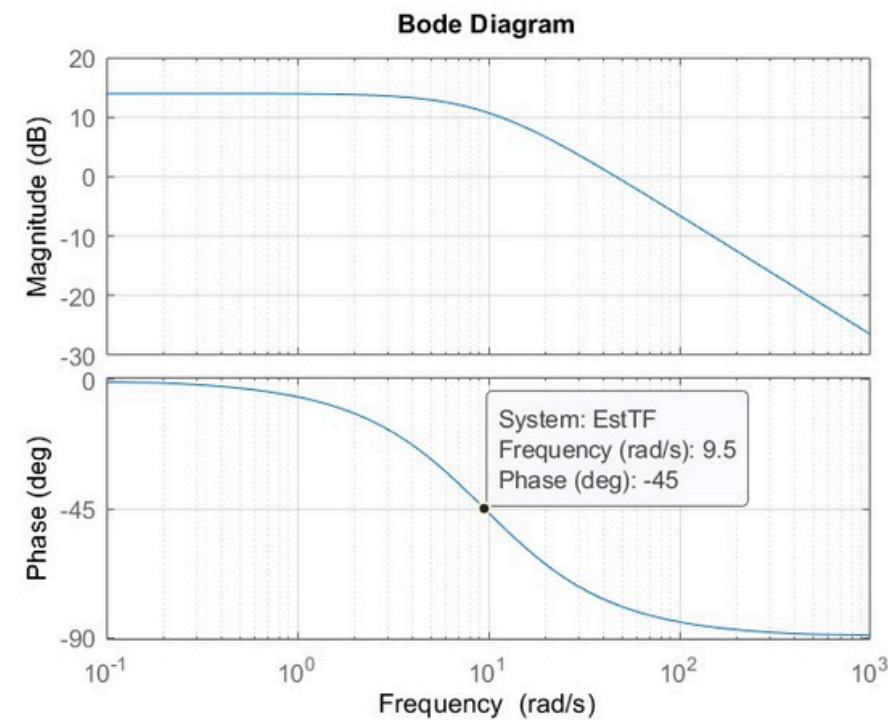
- Input freq: 0.1 ~ 3.0 [Hz]
- Amplitude: Vcmd = 2.0[V] → Vc = 3.3[V]



2차



1차



$$G_{m2}(s) = \frac{790}{s^2 + 16.45s + 172.5}$$

$$G_{m1}(s) = \frac{47.4}{s + 9.515}$$

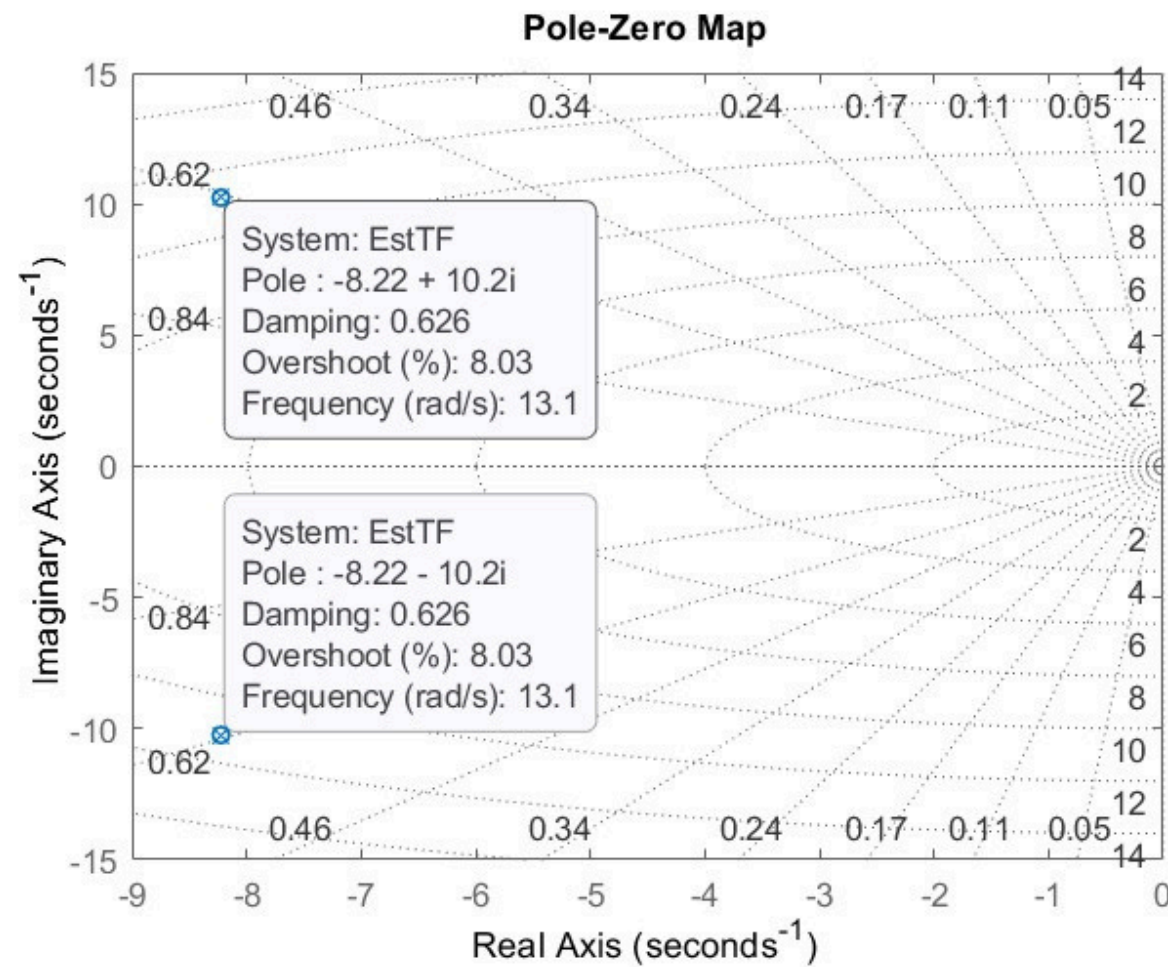
bandwidth = 1.51[Hz]

## 2. Dynamic Motor Modeling

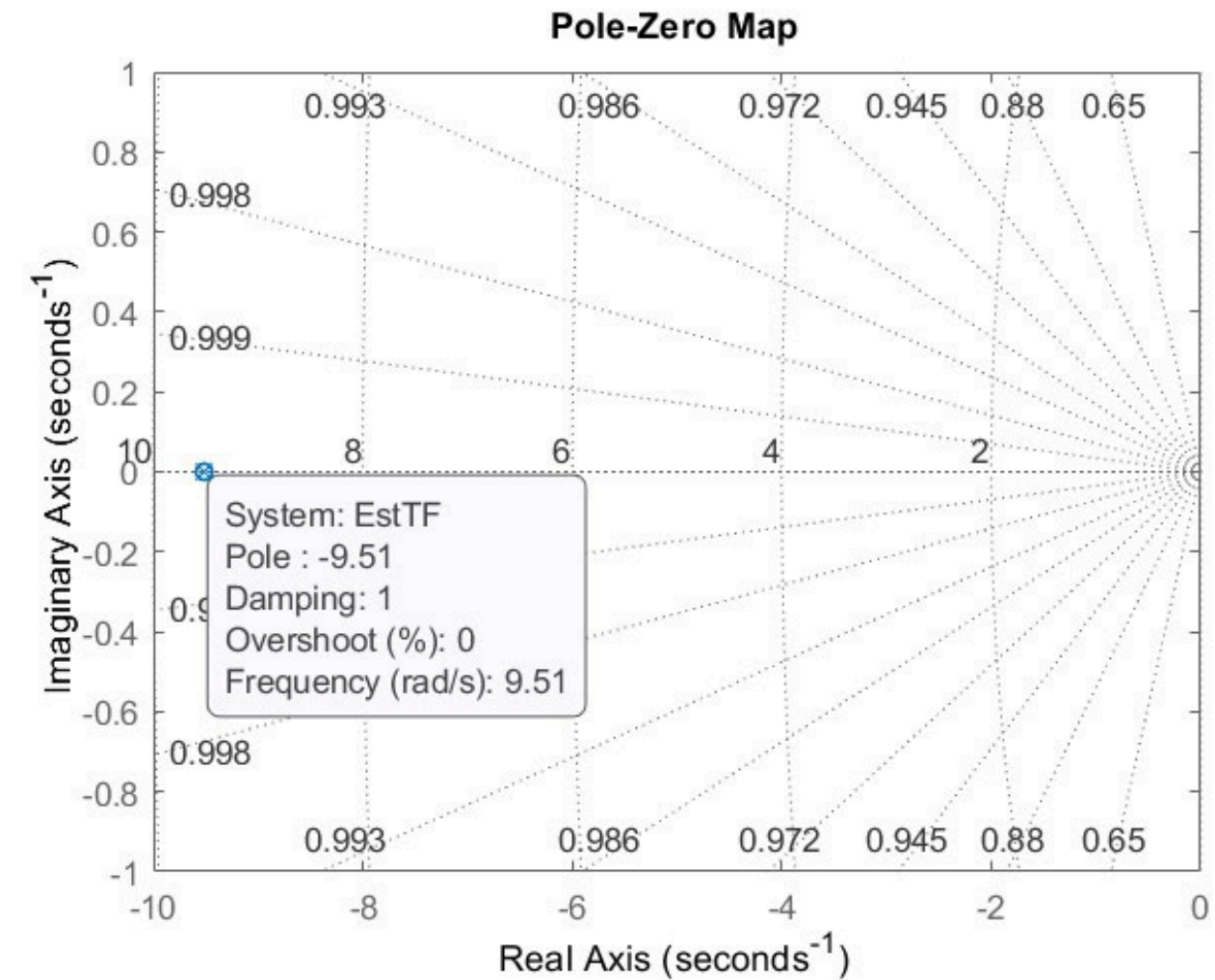
### Pole-zero Map

$$G_m2(s) = \frac{790}{s^2 + 16.45s + 172.5}$$

$$G_m(s) = \frac{47.4}{s + 9.515}$$



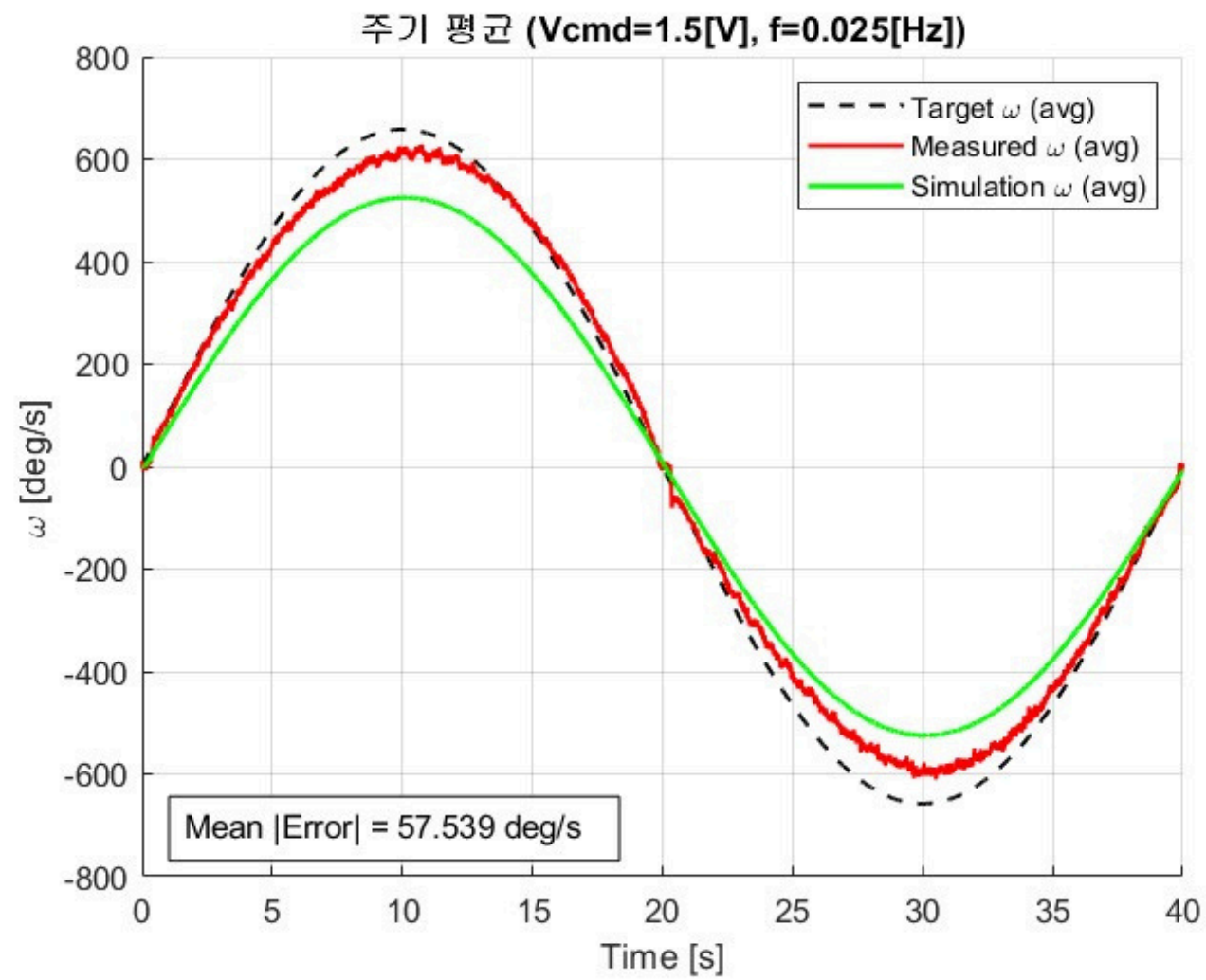
2차



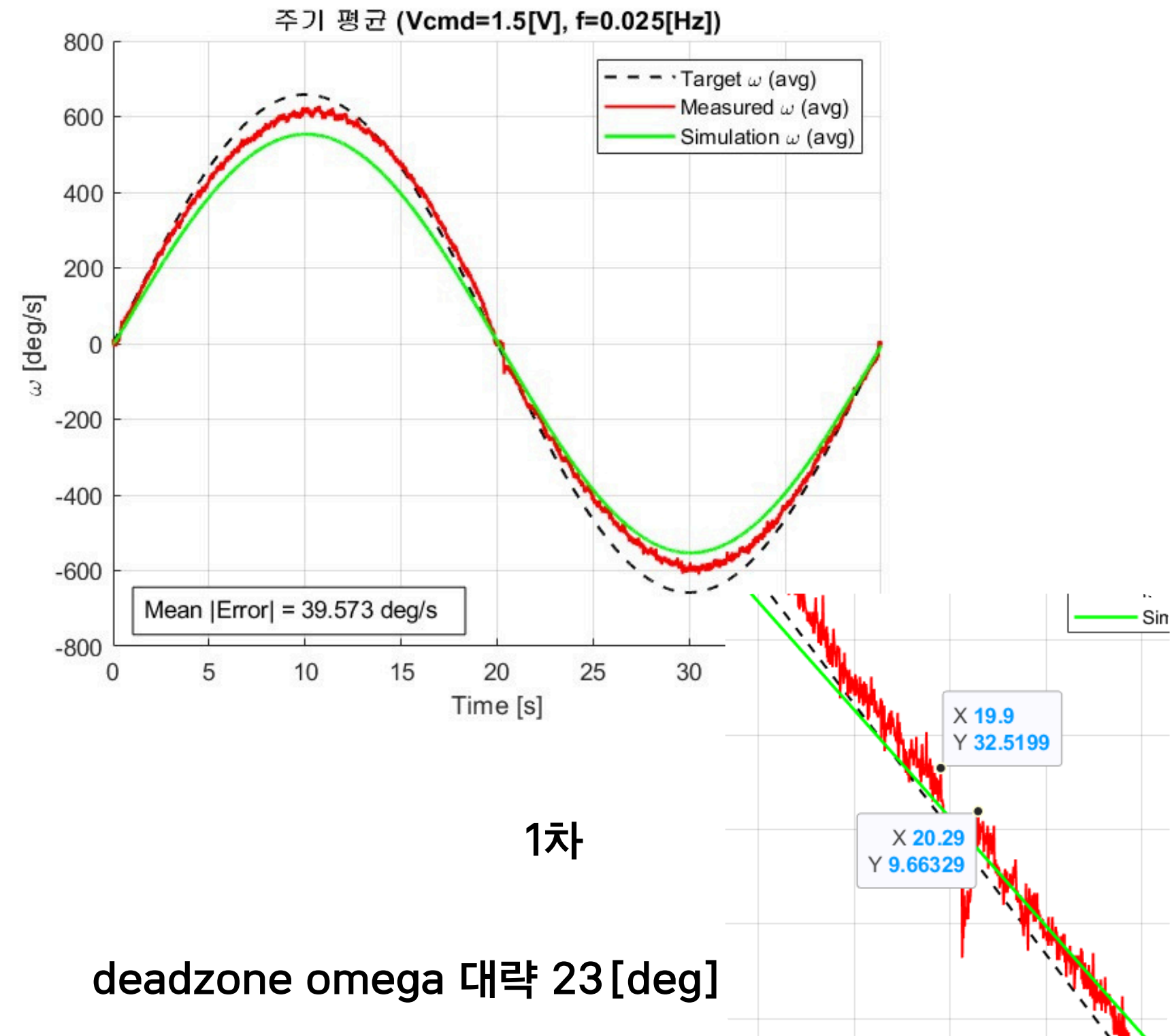
1차

### 3. Model Validation

- 정현파 validation



2차

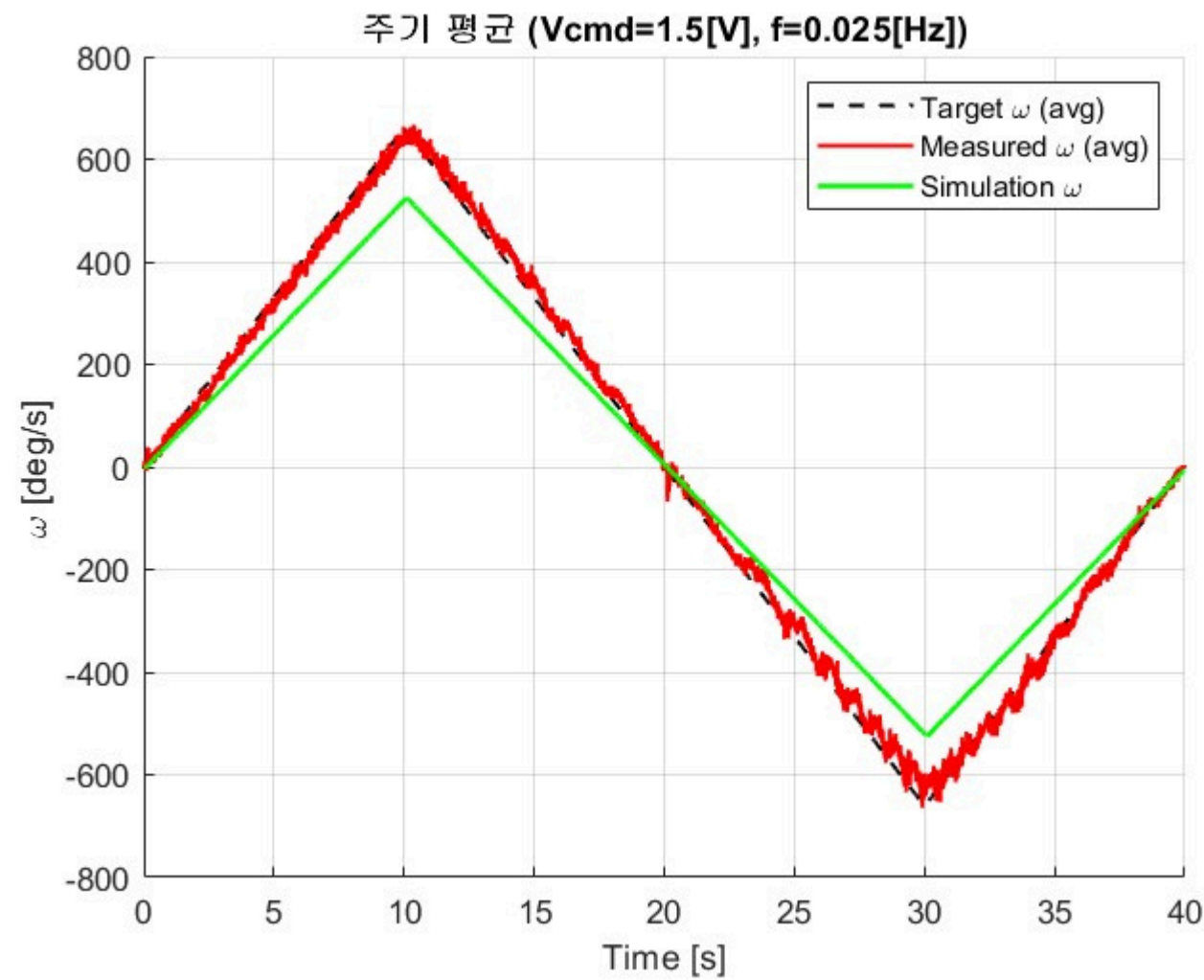


1차

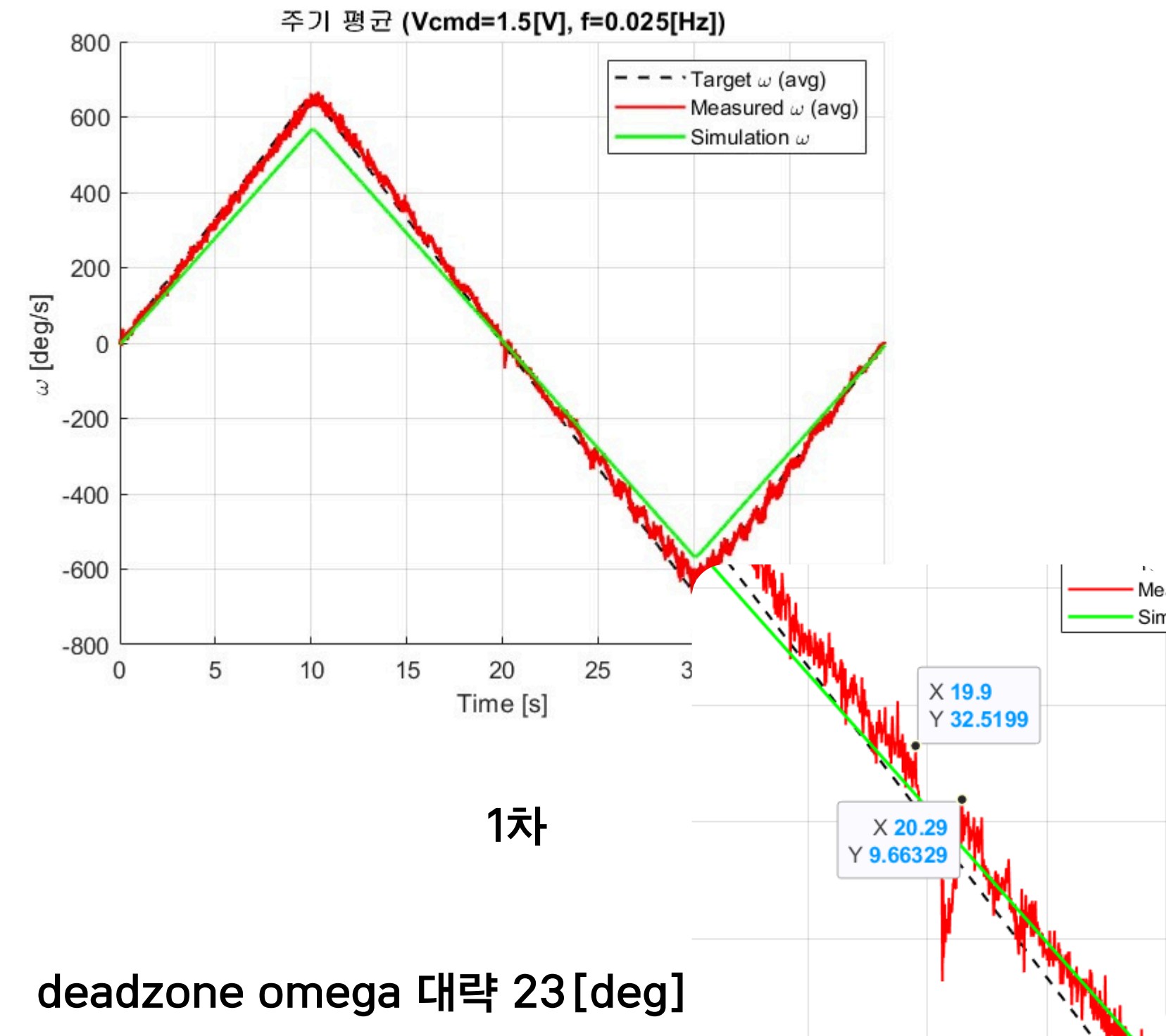
deadzone omega 대략 23[deg]

### 3. Model Validation

- 삼각파 validation



2차



1차

deadzone omega 대략 23[deg]

감사합니다